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Network

TP 2

I. Questions

1. What is the difference between OSI Model and TCP/IP model?

OSI Model: Has 7 layers (physical, data link, network, transport, session, presentation, Application)

TCP/IP Model: has 4 layers (network interface, Internet, Transport, Application)

- The OSI model and TCP/IP model have the same process, but the difference is just the way they represent the name.

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2. How can we manage the users if we have only the information of their IP addresses and Mac addresses?

(Access Control): Access control is the process of authorizing users, groups, and machines to access objects on a network or computer.

Server Administration: Server administration is the process of managing and maintaining computer systems called servers.

- We cannot rely on IP addresses alone to manage users because in a DHCP-based network, IP addresses are assigned dynamically. This means each time a user reconnects, their IP address may change, making it difficult to manage them consistently. So, to manage users, we can assign static IPs to specific machines, but this method is not flexible or efficient for large networks.
- Instead, it's better to manage users based on their MAC addresses, since each device has a unique MAC address that doesn't change. The MAC address can be used in access control lists (ACLs), DHCP reservations, or router configurations to consistently identify and control user access on the network.

3. In the case that you are an administrator of company, and you want to make sure that there is no prohibited programming or application are being used in your organization what should you do?

You can use firewall because the current generation of firewall has application awareness that allow you to block any applications, programs, websites you want to block. Even blocking any content inside any website you want to block. You can also install antivirus or end point protection on all device of the company as rules and regulation.

4. Pick up an open-source monitoring system and install it.

The open-source monitoring system I'm using is Zabbix, on Red Hat Linux. I am using Oracle VirtualBox to create virtual machine and here's the server I'm running:

```
zabbix_appliance-7.4.3 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

AlmaLinux 8.10 (Cerulean Leopard)
Kernel 4.18.0-553.76.1.el8_10.x86_64 on an x86_64

appliance login: root
Password:
*****

Zabbix frontend credentials:

Username: Admin
Password: zabbix

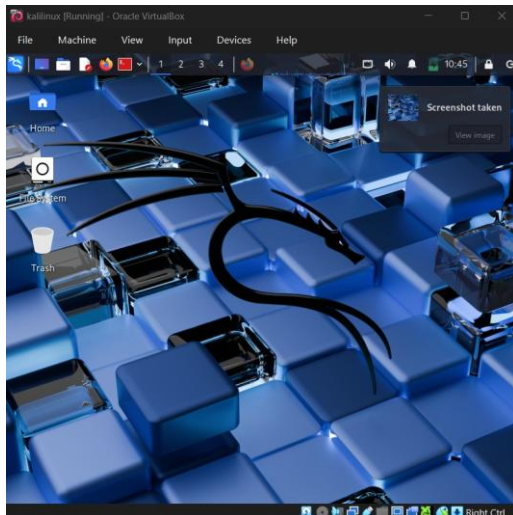
To learn about available professional services, including technical support and training, please visit https://www.zabbix.com/services

Official Zabbix documentation available at https://www.zabbix.com/documentation/current/

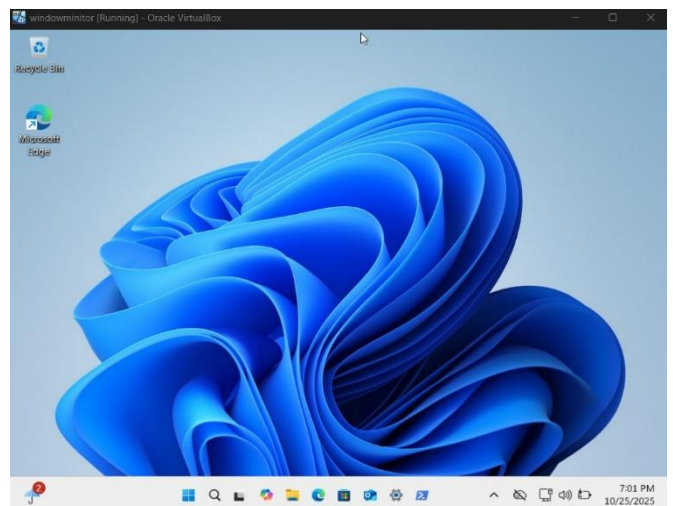
*****

[root@appliance ~]# ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f7:de:c1 brd ff:ff:ff:ff:ff:ff
    altname emp0s3
    inet 192.168.0.14/24 brd 192.168.0.255 scope global dynamic eth0
        valid_lft 28656sec preferred_lft 28656sec
    inet6 fe80::a00:27ff:fcf7:dec1/64 scope link
        valid_lft forever preferred_lft forever
[root@appliance ~]# [ 4535.300832] hrtimer: interrupt took 2294447 ns
```

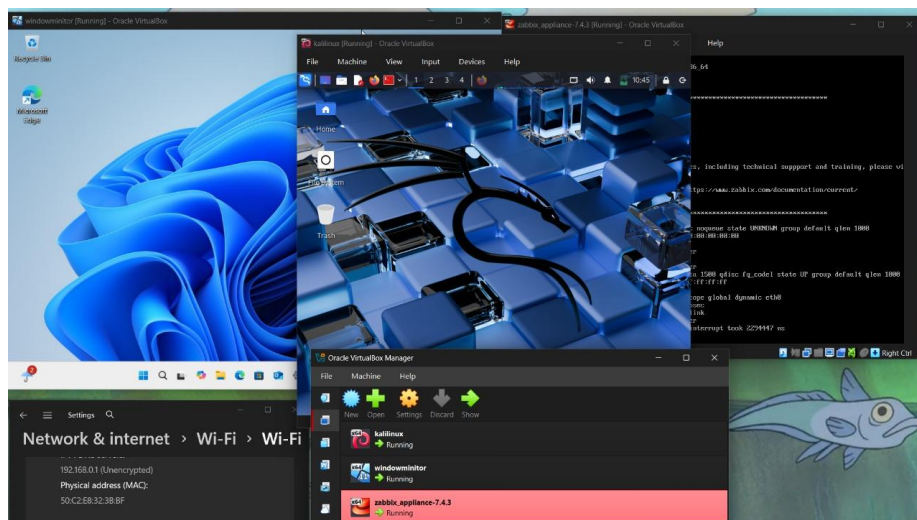
Below here is the two VMs, one is Linux Kali and another one is Windows-11.



Linux-Kali



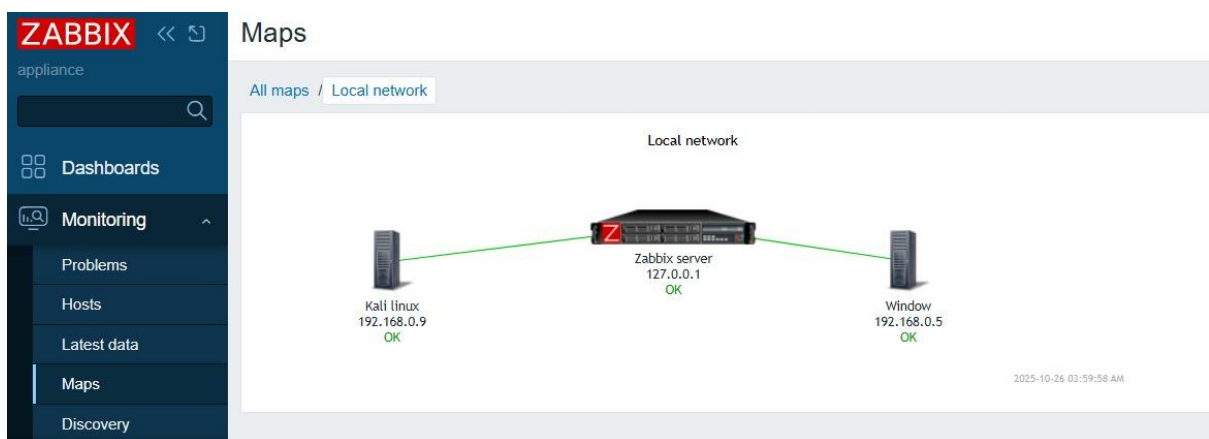
Windows-11



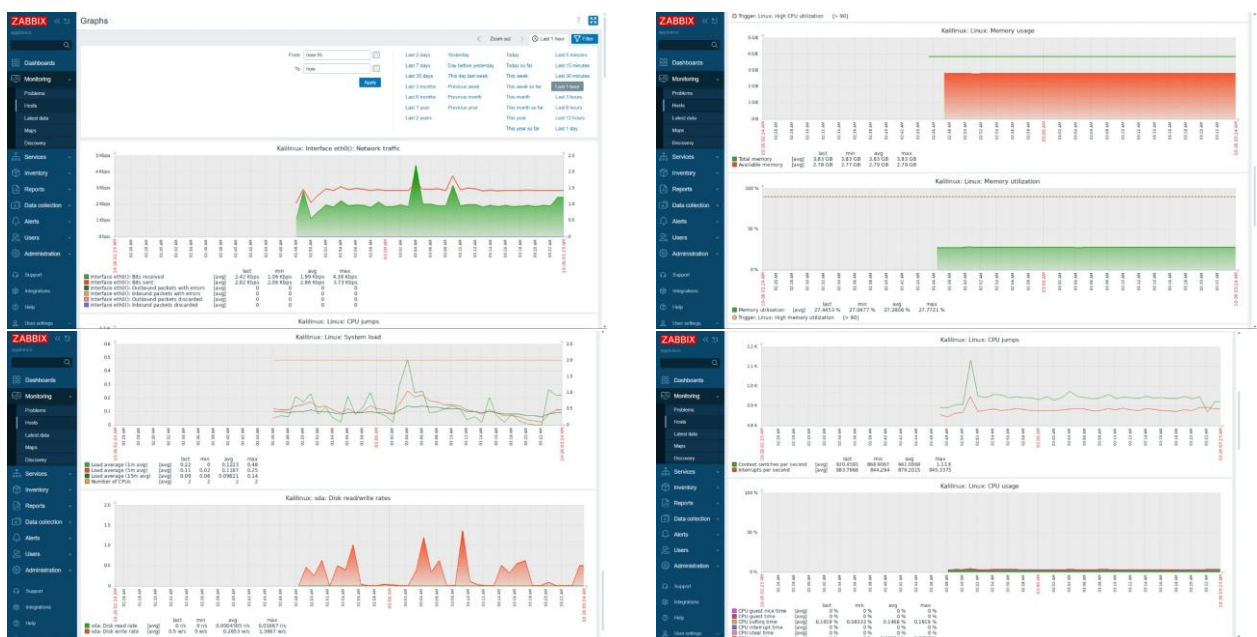
Here is all of them together from my laptop screen and my MAC address

After setting up SNMP on both Windows 11 and Linux Kali, you can go to the Zabbix Frontend (Web) to create host and do connections there, using the Red hat Zabbix server IP address:

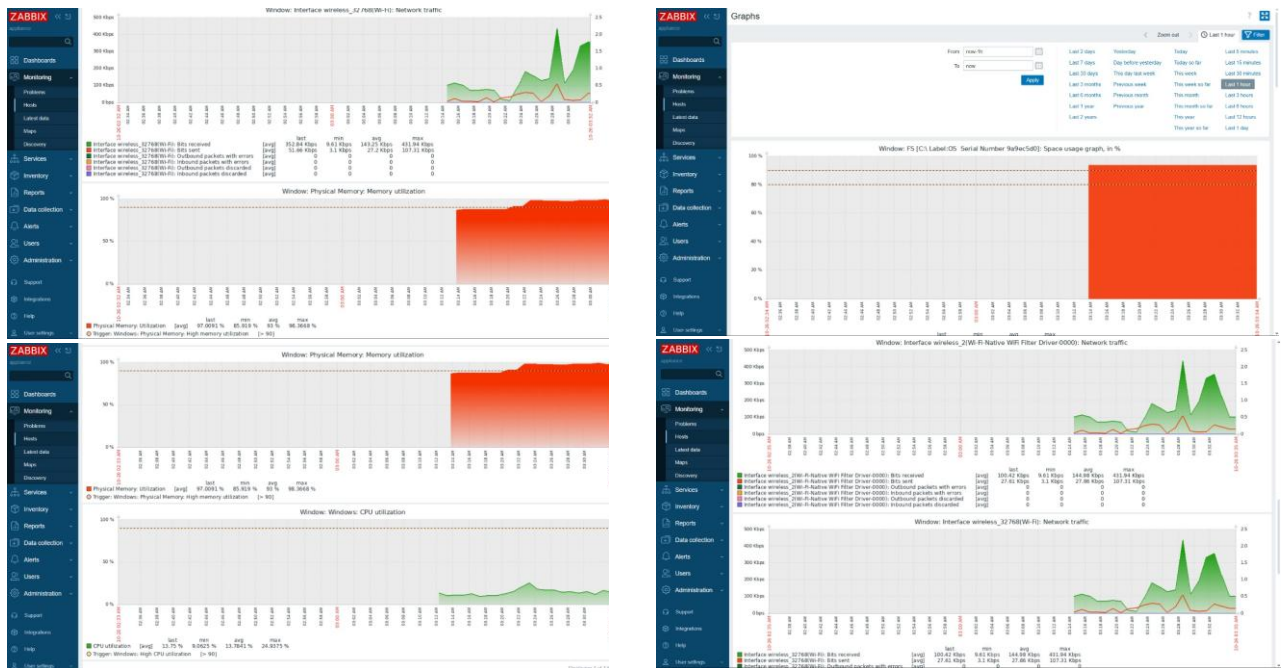
Name	Interface	Availability	Tags	Status	Latest data	Problems	Graphs	Dashboards	Web
Kali linux	192.168.0.9:161	SNMP	class: os target: linux	Enabled	Latest data 57	Problems	Graphs 9	Dashboards 3	Web
Window	192.168.0.5:161	SNMP	class: os target: windows	Enabled	Latest data 50	1	Graphs 7	Dashboards 3	Web
Zabbix server	127.0.0.1:10050	ZBX	class: os class: software subclass: logging	Enabled	Latest data 167	Problems	Graphs 20	Dashboards 4	Web



Here are the data I've selected and collected from the Linux Kali and Windows 11



Linux Kali (Network traffic, Memory, CPU, System load, Disk read/write rate.)



Windows-11 (Network traffic, Memory, CPU, System load, Disk read/write rate)

In reality, there's more to it than meets the eye, this is just the data I've selected, there are plenty more

II. Exercise

1. We need to design new network infrastructure of company (Bank) and that company consists of core network layer, distributed network layer and access network layer.

a. Which components are used in core network layer and its functionality?

→ In core network layer, we use a High-end Core Router or Switch to provide high-speed, reliable, and fault-tolerant switching and routing services to connect the distribution layers. It must have the highest forwarding capacity and minimal latency.

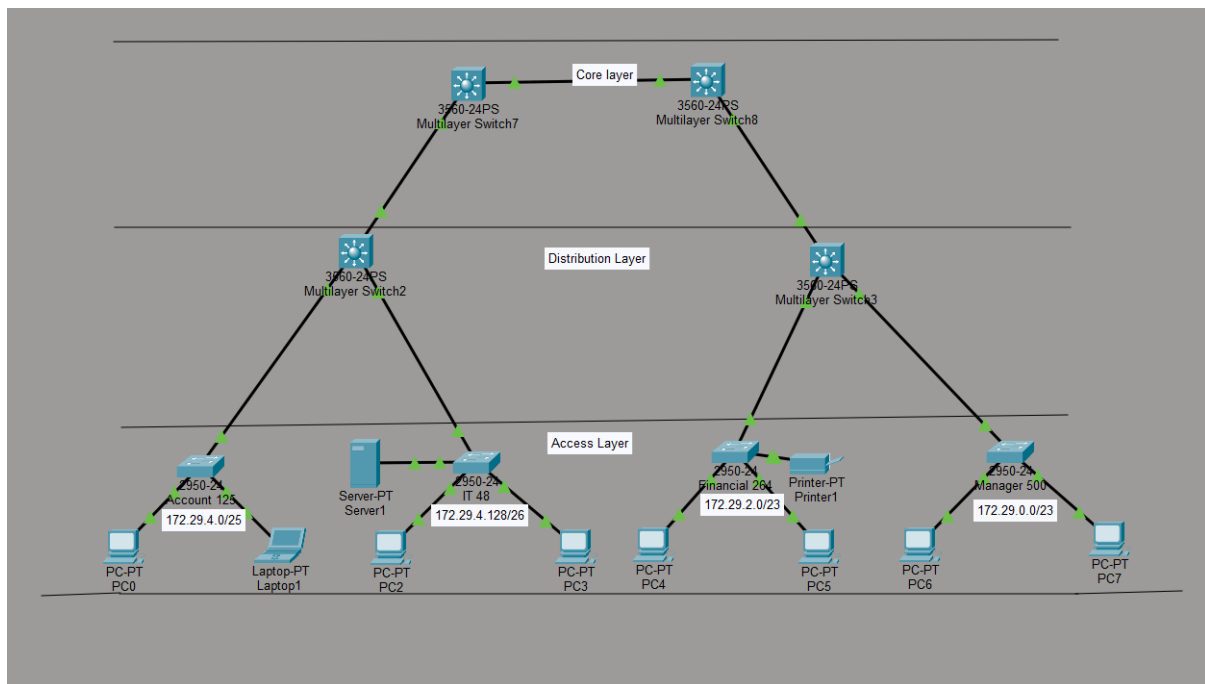
b. Which components are used in distributed network layer and its functionality?

→ In distributed network layer, we also use a switch but its called Layer 3 switch, it acts as a communication point between the access and core layers. It isolates network problems and provides policy-based connectivity.

c. Which components are used in access network layer and its functionality?

→ In access network layer is Layer 2 switches and is where the user end devices is, PCs, Phones, servers, cameras, etc is connected to. It allows for network computing by accessing the user to the network and allow them to store, compute, and transport data.

d. Draw its architecture and apply its IP calculation



Here's the calculation below:

Since we have 4 departments: Manager (500), Financial (264), IT (48), Account (125), we choose class B: 172.29.0.0/16

For the manager department: $500 < 2^9$

Subnetmask = $32 - 9 = 23$

Subnetwork = $2^{mask-16} = 2^{23-16} = 128$

Sub 0: IP_N : 172.29.0.0/23

IP_U : 172.29.0.1/23 → 172.29.1.254/23

IP_B : 172.29.1.255/23

Sub 1: IP_N : 172.29.2.0/23

IP_U : 172.29.2.1/23 → 172.29.3.254/23

IP_B : 172.29.3.255/23

Sub 2: IP_N : 172.29.4.0/23

IP_U : 172.29.4.1/23 → 172.29.5.254/23

IP_B : 172.29.5.255/23

⋮
⋮

Sub 127: IP_N : 172.29.254.0/23

IP_U : 172.29.254.1/23 → 172.29.255.254/23

IP_B : 172.29.255.255/23

For financial department: $264 < 2^9$

Subnetmask = $32 - 9 = 23$

Subnetwork = $2^{mask-23} = 2^{23-23} = 1$

Sub 0: IP_N : 172.29.2.0/23

IP_U : 172.29.2.1/23 → 172.29.3.254/23

IP_B : 172.29.3.255/23

For Accountant department: $125 < 2^7$

Subnetmask = $32 - 7 = 25$

Subnetwork = $2^{mask-23} = 2^{25-23} = 2^2 = 4$

Sub 0: $IP_N: 172.29.4.0/25$

$IP_U: 172.29.4.1/25 \rightarrow 172.29.4.126/25$

$IP_B: 172.29.4.127/25$

Sub 1: $IP_N: 172.29.4.128/25$

$IP_U: 172.29.4.129/25 \rightarrow 172.29.5.254/25$

$IP_B: 172.29.5.255/25$

Sub 2: $IP_N: 172.29.5.0/25$

$IP_U: 172.29.5.1/25 \rightarrow 172.29.6.126/25$

$IP_B: 172.29.6.127/25$

Sub 3: $IP_N: 172.29.5.128/25$

$IP_U: 172.29.5.129/25 \rightarrow 172.29.7.254/25$

$IP_B: 172.29.7.255/25$

For department IT:

$D = 48 < 2^6 = 64$

Subnetmask = $32 - 6 = 26$

Subnetwork = $2^{mask-25} = 2^{26-25} = 2^1 = 2$

Sub 0: $IP_N: 172.29.4.128/26$

$IP_U: 172.29.4.129/26 \rightarrow 172.29.4.190/26$

$IP_B: 172.29.4.191/26$

Sub 1: $IP_N: 172.29.4.192/26$

$IP_U: 172.29.4.193/26 \rightarrow 172.29.4.254/26$

$IP_B: 172.29.4.255/26$

In summary, the IP network of each is:

Manager: $IP_N: 172.29.0.0/23$

Financial: $IP_N: 172.29.2.0/23$

Account: $IP_N: 172.29.4.0/25$

IT: $IP_N: 172.29.4.128/26$